

Smart Waste Collection and Recycling Management System

Pseudo code

START

Create an empty list called waste_records

REPEAT

Display Menu

1. Add Record
2. Display Records
3. Search Record
4. Total Waste Analysis
5. Save Records
6. Load Records
7. Exit

Read user choice

IF choice = 1 THEN

Read House Number

Read Waste Type

Read Quantity

Create tuple:

record $\leftarrow$ (House Number, Waste Type, Quantity)

Add record to waste_records

Display "Record Added Successfully"

ELSE IF choice = 2 THEN

IF waste_records is empty THEN

Display "No Records Available"

ELSE

FOR each record in waste_records DO

Display House Number

Display Waste Type

Display Quantity

END FOR

END IF

ELSE IF choice = 3 THEN

Read House Number to search

FOR each record in waste_records DO

IF House Number matches THEN

Display Record Details

Stop Search

END IF

END FOR

If record not found

Display "Record Not Found"

ELSE IF choice = 4 THEN

total $\leftarrow$ 0

FOR each record in waste_records DO

total $\leftarrow$ total + Quantity

END FOR

Display Total Waste Collected

ELSE IF choice = 5 THEN

Open file "waste_data.txt" in write mode

FOR each record in waste_records DO

Write House Number,

Waste Type,

Quantity into file

END FOR

Close file

Display "Records Saved Successfully"

ELSE IF choice = 6 THEN

Open file "waste_data.txt" in read mode

Read each line from file

Convert data into tuple

Store tuple in waste_records

Close file

Display "Records Loaded Successfully"

ELSE IF choice = 7 THEN

Display "Thank You"

EXIT LOOP

ELSE

Display "Invalid Choice"

END IF

UNTIL choice = 7

STOP

Module-wise Pseudocode

Add record

Input House Number

Input Waste Type

Input Quantity

Create Tuple

Store Tuple in waste_records

Search record

Input House Number

Compare with each record

If found display details

Else display Record Not Found

Save record

Open file

Write all tuples into file

Close file

Waste analysis

Initialize total = 0

Add quantity from each tuple

Display total waste

Source code

```
import customtkinter as ctk

from tkinter import messagebox


ctk.set_appearance_mode("light")
ctk.set_default_color_theme("blue")


# Store records as tuples

waste_records = []


# ----- FUNCTIONS ----- #


def add_record():

    house = house_entry.get().strip()

    waste_type = waste_entry.get().strip().title()

    quantity = quantity_entry.get().strip()


    if house == "" or waste_type == "" or quantity == "":

        messagebox.showerror("Error", "Please fill all fields")

        return


    try:

        quantity = float(quantity)

    except ValueError:

        messagebox.showerror("Error", "Quantity must be a number")
```

```
return
```

```
record = (house, waste_type, quantity)
```

```
waste_records.append(record)
```

```
messagebox.showinfo("Success", "Record Added Successfully")
```

```
house_entry.delete(0, "end")
```

```
waste_entry.delete(0, "end")
```

```
quantity_entry.delete(0, "end")
```

```
display_records()
```

```
def display_records():
```

```
output_box.delete("1.0", "end")
```

```
output_box.insert(
```

```
    "end",
```

```
    "House No\tWaste Type\tQuantity(kg)\n"
```

```
)
```

```
output_box.insert(
```

```
    "end",
```

```
    "-----\n"
```

```
)
```



```
for record in waste_records:
    output_box.insert(
        "end",
        f"{record[0]}\t\t{record[1]}\t\t{record[2]}\n"
    )
```

```
def search_record():
```

```
    house = house_entry.get().strip()
```

```
    for record in waste_records:
```

```
        if record[0] == house:
```

```
            messagebox.showinfo(
                "Record Found",
                f"House Number : {record[0]}\n"
                f"Waste Type : {record[1]}\n"
                f"Quantity : {record[2]} kg"
            )
            return
```

```
messagebox.showerror("Not Found", "Record Not Found")
```

```
def total_waste():
```

```
    total = 0
```

```
    for record in waste_records:
```

```
        total += record[2]
```

```
    messagebox.showinfo(
```

```
        "Waste Analysis",
```

```
        f"Total Waste Collected = {total} kg"
```

```
    )
```

```
def save_records():
```

```
    with open("waste_data.txt", "w") as file:
```

```
        for record in waste_records:
```

```
            file.write(
```

```
                f"{record[0]},{record[1]},{record[2]}\n"
```

```
            )
```

```
    messagebox.showinfo(
```

```
        "Saved",
```

```
    "Records Saved Successfully"  
)
```

```
def load_records():
```

```
    waste_records.clear()
```

```
    try:
```

```
        with open("waste_data.txt", "r") as file:
```

```
            for line in file:
```

```
                house, waste_type, quantity = line.strip().split(",")
```

```
                record = (
```

```
                    house,
```

```
                    waste_type,
```

```
                    float(quantity)
```

```
                )
```

```
                waste_records.append(record)
```

```
    display_records()
```

```
messagebox.showinfo(  
    "Loaded",  
    "Records Loaded Successfully"  
)
```

```
except FileNotFoundError:
```

```
    messagebox.showerror(  
        "Error",  
        "No Saved File Found"  
    )
```

```
# ----- GUI ----- #
```

```
app = ctk.CTk()  
app.geometry("850x700")  
app.title("Smart Waste Collection System")
```

```
title = ctk.CTkLabel(  
    app,  
    text="Smart Waste Collection and Recycling Management System",  
    font=("Arial", 24, "bold")  
)  
title.pack(pady=20)
```

```
# House Number Label
```

```
house_label = ctk.CTkLabel(  
    app,  
    text="House Number",  
    font=("Arial", 16)  
)  
house_label.pack()
```

```
house_entry = ctk.CTkEntry(  
    app,  
    width=300  
)  
house_entry.pack(pady=10)
```

```
# Waste Type Label  
waste_label = ctk.CTkLabel(  
    app,  
    text="Waste Type",  
    font=("Arial", 16)  
)  
waste_label.pack()
```

```
waste_entry = ctk.CTkEntry(  
    app,  
    width=300  
)  
waste_entry.pack(pady=10)
```

```
# Quantity Label
```

```
quantity_label = ctk.CTkLabel(  
    app,  
    text="Quantity (kg)",  
    font=("Arial", 16)  
)  
quantity_label.pack()
```

```
quantity_entry = ctk.CTkEntry(  
    app,  
    width=300  
)  
quantity_entry.pack(pady=10)
```

```
# Buttons
```

```
ctk.CTkButton(  
    app,  
    text="Add Record",  
    command=add_record,  
    width=200  
)
```

```
ctk.CTkButton(  
    app,  
    text="Display Records",
```

```
        command=display_records,  
        width=200  
    ).pack(pady=5)
```

```
    ctk.CTkButton(  
        app,  
        text="Search Record",  
        command=search_record,  
        width=200  
    ).pack(pady=5)
```

```
    ctk.CTkButton(  
        app,  
        text="Total Waste Analysis",  
        command=total_waste,  
        width=200  
    ).pack(pady=5)
```

```
    ctk.CTkButton(  
        app,  
        text="Save Records",  
        command=save_records,  
        width=200  
    ).pack(pady=5)
```

```
    ctk.CTkButton(  
        app,  
        text="Delete Records",  
        command=delete_records,  
        width=200  
    ).pack(pady=5)
```

```
    app,  
    text="Load Records",  
    command=load_records,  
    width=200  
).pack(pady=5)
```

```
# Output Area
```

```
output_box = ctk.CTkTextbox(  
    app,  
    width=700,  
    height=250  
)  
output_box.pack(pady=20)
```

```
app.mainloop()
```


Output screenshot

```
python assignment project.py - C:/Users/smohi/AppData/Local/Programs/Python/Python312/python assignment project.py (3.12.7)
File Edit Format Run Options Window Help
from tkinter import messagebox

ctk.set_appearance_mode("light")
ctk.set_default_color_theme("blue")

# Store records as tuples
waste_records = []

# ----- FUNCTIONS ----- #

def add_record():
    house = house_entry.get().strip()
    waste_type = waste_entry.get().strip().title()
    quantity = quantity_entry.get().strip()

    if house == "" or waste_type == "" or quantity == "":
        messagebox.showerror("Error", "Please fill all fields")
        return

    try:
        quantity = float(quantity)
    except ValueError:
        messagebox.showerror("Error", "Quantity must be a number")
        return

    record = (house, waste_type, quantity)
    waste_records.append(record)

    messagebox.showinfo("Success", "Record Added Successfully")

    house_entry.delete(0, "end")
    waste_entry.delete(0, "end")
    quantity_entry.delete(0, "end")

    display_records()

def display_records():
    output_box.delete("1.0", "end")

    output_box.insert(
        "end",
        "House No\tWaste Type\tQuantity(kg)\n"
    )
    output_box.insert(
        "end",
        "-----\n"
    )
)
```

Smart Waste Collection System

Smart Waste Collection and Recycling Management System

Add Record

Display Records

Search Record

Total Waste Analysis

Save Records

Load Records

Record adding

Smart Waste Collection System

Smart Waste Collection and Recycling Management System

House Number
4

Waste Type
wool

Quantity (kg)
5

Add Record
Display Records
Search Record
Total Waste Analysis
Save Records
Load Records

Display Records

Success

Record Added Successfully

OK

House No	Waste Type	Quantity(kg)
4	Wool	5.0

Smart Waste Collection and Recycling Management System

House Number

Waste Type

Quantity (kg)

Add Record

Display Records

Search Record

Total Waste Analysis

Save Records

Load Records

House No Waste Type Quantity(kg)

4	Wool	5.0
34	Rubber	8.0
40	Paper	10.0
12	Wool	3.0

House No Waste Type Quantity(kg)

4	Wool	5.0
34	Rubber	8.0
40	Paper	10.0
12	Wool	3.0
6	Plastic	23.0

Display record

Smart Waste Collection and Recycling Management System

House Number

Waste Type

Quantity (kg)

Add Record

Display Records

Search Record

Total Waste Analysis

Save Records

Load Records

House No	Waste Type	Quantity(kg)
4	Wool	5.0
34	Rubber	8.0
40	Paper	10.0
12	Wool	3.0
6	Plastic	23.0

Search record

House Number

6

Waste Type

Quantity (kg)

Add Record

Display Records

Search Record

Total Waste Analysis

Save Records

Load Records

Record Found

i

House Number : 6
Waste Type : Plastic
Quantity : 23.0 kg

OK

House Number

1

Waste Type

Quantity (kg)

Add Record

Display Records

Not Found

Record Not Found

OK

Total waste analysis in a single day

Waste Analysis

Total Waste Collected = 49.0 kg

OK

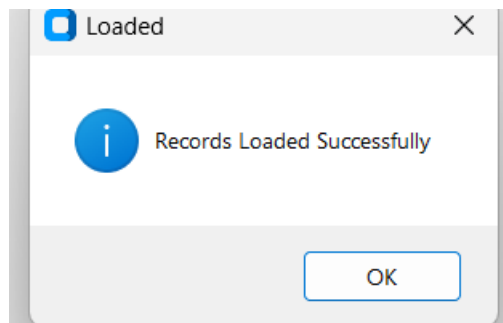
Load record

Smart Waste Collection and Recycling Management System

House Number

Waste Type

Quantity (kg)



House No	Waste Type	Quantity(kg)
4	Wool	5.0
34	Rubber	8.0
40	Paper	10.0
12	Wool	3.0
6	Plastic	23.0

Save record

House Number

Waste Type

Quantity (kg)

Add Record

Display Records

Saved

i

Records Saved Successfully

OK

